

Inputs

Weather: ERA5 2025;
ECMWF day-ahead
forecasts

Workload traces
Azure LLM,
Alibaba jobs

Tariffs, prices
ADDC, DEWA,
ERCOT

System signal
synthetic 2030
net load

Metered buildings
BDG2, three
hot U.S. sites

Operation (RQ1, RQ2)

Facility LP
deferral, $COP(\theta)$,
TES, battery, PV

Scenarios S0–S4, R
perfect foresight
(H1–H3)

MPC from forecasts
expected price,
margin (H6)

Dispatch P_t^g
firm reduction $F(d)$,
value Γ

Settlement (RQ3)

Baselines B1–B5
B5 physics-guided,
pretrained (H5)

Commit before event
 $C = H(\hat{B} \parallel h_M \parallel$
 $h_X \parallel \tau_c \parallel n)$

Reveal, verify
and settle
 $\Sigma(\hat{B}_t - P_t^g)$

Tests: placebo, inflation,
shopping (H4); real
buildings (H5)

validation